

Lecture 14 Feb 18th

Hydrogen atom:

$$\hat{H} = \frac{\hat{p}^2}{2m} - \frac{\alpha}{r};$$

Natural units $c = 1; \hbar = 1$

$$\left\{ \begin{array}{l} a_0 = \frac{1}{\alpha m_e}; \quad E_n^{(10)} = -\frac{\alpha^2 m_e}{2n^2} = -\frac{\alpha}{2} \frac{1}{a_0 n^2} \\ \alpha = \frac{1}{137}; \quad m_e = 511 \text{ keV} \approx 0.5 \text{ MeV} \quad \frac{\hat{p}^2}{2m n^2} \\ \frac{1}{a_0} \approx \text{few keV} \rightarrow \text{atomic physics} \end{array} \right.$$

$$\Psi_{nlm} = R_{nl}(r) Y_{lm}(\theta, \phi)$$

$$\rho = \frac{r}{na_0}$$

$$= C \times e^{-\rho/2} \rho^l \underbrace{P_{n-l-1}^{2l+1}(\rho)}_{\text{"modes"}} Y_l^m(\theta, \phi)$$

"modes" $\rightarrow$ Polynomial of deg. $N = n - (l+1)$

	s ($\ell = 0$)	p ($\ell = 1$)				d ($\ell = 2$)					f ($\ell = 3$)							
	$m = 0$	$m = 0$	$m = \pm 1$		$m = 0$	$m = \pm 1$		$m = \pm 2$		$m = 0$	$m = \pm 1$		$m = \pm 2$		$m = \pm 3$			
	s	p _z	p _x	p _y	d _{z²}	d _{xz}	d _{yz}	d _{xy}	d _{x²-y²}	f _{z³}	f _{xz²}	f _{yz²}	f _{xyz}	f _{z(x²-y²)}	f _{x(x²-3y²)}	f _{y(3x²-y²)}		
$n = 1$																		
$n = 2$																		
$n = 3$																		
$n = 4$																		
$n = 5$										...	...	...	...	...	...	...	...	
$n = 6$					...†	...†	...†	...†	...†	...*	...*	...*	...*	...*	...*	...*	...*	
$n = 7$		...†	...†	...†	...*	...*	...*	...*	...*	...*	...*	...*	...*	...*	...*	...*	...*	

Nodes

$$N = n - l - 1$$

$$l = 0, \dots, n-1 \rightarrow n$$

$$m = -l, \dots, l \rightarrow \underline{2l+1}$$

Degenerate states:

$$\sum_{l=0}^{n-1} (2l+1) = n^2$$

Represented by $|n l m\rangle$

$$\langle \hat{r} | 100 \rangle = \frac{e^{-r/a_0}}{\sqrt{\pi a_0^3}}$$

BTW: $\langle \hat{r} | \hat{p} | 100 \rangle = -i \nabla \Psi_{100}(r) = -i \partial_r \Psi_{100}(r)$
 $= \frac{i}{a_0} \Psi_{100}(r)$

Typically $\rho \sim \frac{1}{a_0} \sim \alpha m_e \doteq m_e v$

$\Rightarrow v \approx \alpha \rightarrow$ non-Rel

Does $v \rightarrow 1$ when $Z\alpha \rightarrow 1$?

Missing quantum number: m_s e spin!

$(s, m_s): s = \frac{1}{2}; m_s = \pm \frac{1}{2}$

See H:

$$\hat{L}^2 |n l m\rangle = l(l+1) |n l m\rangle$$

$$\hat{L}_z |n l m\rangle = m |n l m\rangle \quad \text{3/4}$$

Enhance to $\hat{S}^2 |n l m_l m_s\rangle = \frac{1}{2}(\frac{1}{2}+1) |n l m_l m_s\rangle$

Uncoupled basis $\rightarrow \hat{L} \cdot \hat{S}$? ^{Spin} ^{Orbit.}

Now, the problem:

Suppose we have a perturbation that depends on the total angular momentum:

$$\hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}}$$

$$\hat{\mathbf{J}}^2 |j m_j\rangle = j(j+1) |j m_j\rangle$$

$$\hat{J}_z |j m_j\rangle = m_j |j m_j\rangle,$$

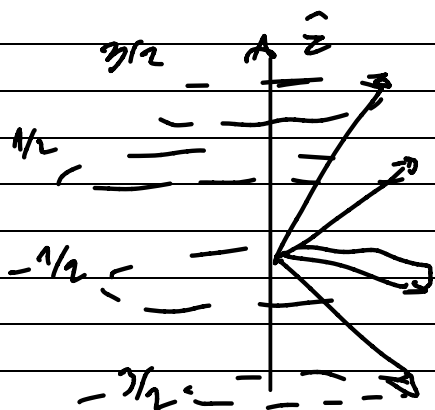
This is a better basis. But

how is it related to $|n l m_l m_s\rangle$?

Note that $\hat{J}_z = \hat{L}_z + \hat{S}_z$

$$[\hat{\mathbf{J}}^2, \hat{J}_z] = 0; \quad [\hat{J}_i, \hat{J}_j] = i \epsilon_{ijk} \hat{J}_k$$

$$\epsilon_{xyz} = +1$$



$$\left\{ \begin{array}{l} \mathcal{H} \approx \mathcal{H}_L \otimes \mathcal{H}_S \\ \dim(\mathcal{H}_L) = 2l+1 \\ \dim(\mathcal{H}_S) = 2 \end{array} \right.$$

$$|n l m_l "s" m_s\rangle = |n l m_l m_s\rangle$$

$$\hat{L}^2, \hat{L}_z, \hat{S}^2, \hat{S}_z.$$

I want to find the basis that diagonalizes $\hat{J}$.

$$\hat{L}^2, \hat{S}^2, \hat{J}^2, \hat{J}_z$$

$$|n l "s" j m_j\rangle = |n l j m_j\rangle$$

⚠ Addition of angular momentum: SU(2)

$$\begin{cases} (2l+1) \otimes 2 = (2l+2) \oplus (2l) & \text{dimension} \\ \text{or} \\ l \otimes \frac{1}{2} = (l+\frac{1}{2}) \oplus (l-\frac{1}{2}) & \text{spin label} \end{cases}$$

BTW:

$$\bullet l=0 \begin{cases} 1 \otimes 2 = 2 \oplus 0 = 2 \\ 0 \otimes \frac{1}{2} = \frac{1}{2} \end{cases}$$

$S_{1/2}$

$$\bullet l=1 \begin{cases} \text{Reducible} & \text{Irreducible} \\ 3 \otimes 2 = 4 \oplus 2 \\ 1 \otimes \frac{1}{2} = \frac{3}{2} \oplus \frac{1}{2} \end{cases}$$

$P_{3/2} \oplus P_{1/2}$

$$\bullet l=2 \begin{cases} 5 \otimes 2 = 6 \oplus 4 \\ 2 \otimes \frac{1}{2} = \frac{5}{2} \oplus \frac{3}{2} \end{cases}$$

$D_{5/2} \oplus D_{3/2}$

$\vdots$

$\vdots$

	S $l=0$	P $l=1$	D $l=2$
$n=4$	<u>4 $S_{1/2}$</u>	<u>4 $P_{3/2}$</u> 4 $P_{1/2}$	<u>4 $D_{3/2}$</u> 4 $D_{5/2}$
$n=3$	<u>3 $S_{1/2}$</u>	<u>3 $P_{3/2}$</u> 3 $P_{1/2}$	<u>3 $D_{3/2}$</u> 3 $D_{5/2}$
$n=2$	<u>2 $S_{1/2}$</u>	<u>2 $P_{3/2}$</u> 2 $P_{1/2}$	
$n=1$	<u>1 $S_{1/2}$</u>		

always $\frac{1}{2}$

because

$$1 \otimes 2 = 2$$

always $\frac{3}{2} + \frac{3}{4}$

because

$$3 \otimes 2 = 4 \oplus 2$$

change of basis:

$$|n \ell j m_j\rangle = \sum_{m_\ell m_s} |n \ell m_\ell m_s\rangle \underbrace{\langle n \ell m_\ell m_s | j m_j \rangle}_{C_{\ell m_\ell, s m_s}^{j m_j}} \quad \delta m_j; m_\ell + m_s$$

Note that $m_j = m_s + m_\ell$

Proof:

$$\hat{J}_z | \ell m_\ell m_s \rangle = (S_z + L_z) | \ell m_\ell m_s \rangle$$
$$= (m_\ell + m_s) | \ell m_\ell m_s \rangle$$

$$\hat{J}_z | \ell j m_j \rangle = m_j | \ell j m_j \rangle$$

$$\Rightarrow \boxed{\langle \ell m_\ell m_s | \hat{J}_z | \ell j m_j \rangle = m_j C_{\ell m_\ell, s m_s}^{j m_j}}$$
$$= (m_\ell + m_s) C_{\ell m_\ell, s m_s}^{j m_j}$$

$$\Rightarrow m_j = m_\ell + m_s.$$

Perturbation

Spin - Orbit Coupling

$$H = H_0^{\text{FS}} + \frac{\alpha^2}{2} j(r) \hat{L} \cdot \hat{S}$$

$$= \frac{1}{2m} \left[\hat{p}_r^2 + \frac{\hat{L}^2}{r^2} \right] + V(r) + \frac{\alpha^2}{2} j(r) \hat{L} \cdot \hat{S}$$

$$\hat{p}^2 = \hat{p}_r^2 + \frac{\hat{L}^2}{r^2}$$

$$\text{recall: } \left[-\hbar^2 \frac{d^2}{dr^2} + \frac{\hbar^2 l(l+1)}{r^2} \right] \psi_{nlm}$$

$$\hat{J}^2 = \hat{L}^2 + \hat{S}^2 + 2\hat{L} \cdot \hat{S} \Rightarrow \hat{L} \cdot \hat{S} = \frac{\hat{J}^2 - \hat{L}^2 - \hat{S}^2}{2}$$

$$\begin{aligned} \hat{L} \cdot \hat{S} |l j m_j\rangle &= \frac{1}{2} (\hat{J}^2 - \hat{L}^2 - \hat{S}^2) |l j m_j\rangle \\ &= \frac{1}{2} \left[j(j+1) - l(l+1) - \overset{3/4}{s(s+1)} \right] |l j m_j\rangle \end{aligned}$$

BTW L^2, S^2, J^2 all commute w/ $(\hat{L} \cdot \hat{S})$ and J^2

$$[L^2, L_i] = 0; [S^2, S_i] = 0 \Rightarrow$$

$$\text{But } [\hat{L} \cdot \hat{S}, L_z] \neq 0$$

$$[\hat{L} \cdot \hat{S}, S_z] \neq 0$$

→ M_s and M_L are not good Quantum numbers.

$$\text{But } [\hat{L} \cdot \hat{S}, J^2] = [L \cdot S, L_z + S_z]$$

$$= \frac{1}{2} ([J^2, J^2] - [L^2, (L_z + S_z)] - [\hat{S}^2, (L_z + S_z)]) = 0$$

⇒ Already Known

$$E_n^{\text{so}} = \langle n l j m_j | \frac{\alpha^2}{2} \delta(r) \hat{L} \cdot \hat{S} | n l j m_j \rangle$$

$$= \frac{\alpha^2}{2} \langle \delta(r) \rangle \left(\frac{j(j+1) - l(l+1) - \frac{3}{4}}{2} \right)$$

$$\vec{\mu}_L = -\frac{e \hbar}{2m_e} \vec{L} \quad \vec{\mu}_S = -\frac{g_e e \hbar}{2m_e} \vec{S}$$

$$V = -\vec{\mu} \cdot \vec{B}$$

$$\hat{V}_L \longrightarrow E_{m_L}^{(1)} = \mu_B B m_L$$

$$\hat{V}_S \longrightarrow E_{m_S}^{(1)} = \mu_B B m_S$$

$\vec{B}_L$ induced by nucleus orbiting electron:

$$\hat{V}_{LS} = -\vec{\mu}_S \cdot \vec{B}_L$$

what is B_L ? $B_L = -\vec{r} \times \vec{E}$

$$= -\frac{1}{e m_e} \left(\frac{1}{r} \frac{dV}{dr} \right) \hat{L}$$

$$\frac{dV}{dr} = + \frac{\alpha Z}{r^2} \Rightarrow B_L = \frac{\alpha Z}{e r^3 m_e}$$

$$\langle g(r) \rangle = \int_0^\infty R_{nl}^2(r) \frac{1}{r} \frac{dV}{dr} r^2 dr$$